

Rhylthyme: A Declarative Language and Agent-Native Runtime for Human-Executed, Resource-Constrained Schedules

Jeremy Leipzig
leipzig@gmail.com

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Abstract

Scheduling software tends to fall into three kinds: tools that compute a plan and hand it to a person as a static artifact; engines that execute a dependency graph on machines with no person in the loop; and checklists with timers that walk a person through steps without any model of shared resources or parallel work. Rhylthyme (“real time”) is built for what falls between them: capacity-constrained, multi-track schedules whose executor is a human working against a clock, with steps the executor starts and ends. We describe (i) a compact JSON language of parallel tracks, timed steps, dependency triggers, environments and resource constraints, annotated with workflow and time ontology terms; (ii) a validator, an auto-planner with several strategies, and two runtimes (a terminal runner and a web player with timeline, itinerary and dependency-graph views); and (iii) a Model Context Protocol server that exposes validation, timing analysis, catalog search, import and publication as annotated tools, so that a language-model agent can author and deliver a schedule as tool calls rather than prose. We position the system as a natural extension of traditional project scheduling techniques, and compare it with air-traffic flow management, temporal constraint networks and robot task scheduling, cooking-step and laboratory schedulers, lab-automation software, and workflow engines, arguing that its contribution is a narrow one: capacity constraints and cross-track dependencies inside a followable, human-executed timeline, one schema used across domains, and an agent interface whose validator refuses malformed programs before they are published.

1 Introduction

Many tasks consist of several manual steps carried out in serial or in parallel, under resource constraints, with durations that are only partly known in advance. A cook coordinating four dishes so that they finish together, a technician running a protocol whose incubations overlap, a stage manager calling cues, and a trainer running superset all face the same structure: several sequential lines of work that run at once, dependencies between them, durations that are sometimes fixed and sometimes open-ended, and a handful of resources (one oven, two thermocyclers, one PA system) that cannot be used by more than a bounded number of steps at a time. Abbott, Bogart and Walkingshaw call the artifacts that describe such work “programs for people”: recipes, protocols and run sheets are written by one person to be executed by another, present an idealized linear process, and rely on conventions for widening or narrowing what a step means [1].

Computers have offered solutions to these logistical timing problems for decades. The Gantt chart, a bar chart with one row per activity and time on the horizontal axis; the critical path method (CPM), which links activities by precedence and finds the chain that determines the finish date; the Program Evaluation and Review Technique (PERT), which does the same with three duration estimates per activity (optimistic, most likely, pessimistic) so that the finish date carries an uncertainty; and resource-constrained project scheduling, which adds limited resources that activities compete for, all produce a plan from a formal description of the work [12, 32, 41, 76]. The obstacle has been the description. Representing a schedule in a machine-readable syntax is too arduous for the people who actually run these tasks, and mixed-initiative scheduling workbenches, which tried to meet users halfway with interactive editing of a formal model, never reached the kitchen or the bench [28, 49]. Before large language models, an end user had no way to state a schedule, its constraints, or how it should be optimized in natural language and have software act on the statement. That has changed: language-model agents now translate free-text protocols into executable form and draft formal planning problems from prose [5, 9, 58], which makes a compact, well-specified target syntax worth having.

Meanwhile, the systems that do handle logistics in real time fall at the extremes. Some are highly specialized and proprietary: airline operations and air-traffic flow management run on bespoke optimization and collaborative decision-making platforms [10, 73]. Some are entirely manual: a bench laboratory typically runs overlapping incubations with an array of kitchen timers and sticky notes, and the protocol platforms that assist it step through a checklist with a single timer [62]. And some are entirely automated: factory scheduling, liquid-handling robots and self-driving laboratories dispatch steps to machines that need no explanation [52, 63, 75]. Between the extremes, the tooling that does put a person in the loop (guided-cooking timers, show control, protocol run modes) lacks the project-scheduling model of parallel tracks, cross-track dependencies and capacity limits on shared resources, while domain schedulers for kitchens and laboratories have the model but stop at an optimized procedure [42, 48, 81]. Workflow engines and laboratory orchestrators execute dependency graphs, but their executors are compute nodes, robots or instruments, and their objective is reproducible output rather than a followable timeline [39, 63].

Rhylthyme is a small system built for the intersection. Its contributions are:

1. **A language.** A JSON program model of parallel *tracks* of sequential *steps*, with a fixed vocabulary of start triggers, three kinds of duration, buffers, choice branches, replicates with staggered starts, and a separate *environment* document that declares resource capacities and the people available to attend them. The schema annotates its constructs with workflow and time ontology terms, so that each has a stated meaning in a standard temporal model.
2. **A runtime.** A validator that catches structural and scheduling errors, an auto-planner with named strategies (finish together, minimize length, provide consistent activity coverage, fit to a target time) and over-utilization detection, a curses runner with manual and indefinite steps, actor accounting and abort handling, and a web application whose timeline, itinerary and dependency-graph views play the schedule.
3. **An agent interface.** A Model Context Protocol (MCP) server that exposes `validate_program`, `analyze_schedule`, `visualize_schedule`, import from external recipe and protocol sources, catalog search and account tools, with tool annotations, structured output, resources and a prompt. One implementation serves five endpoints (generic, kitchen, laboratory, events, gym) from one schema.

2 Related work

Gantt charts, CPM and PERT. The Gantt chart remains the dominant visual grammar for “which activity runs when on which row” [76], and Rhylthyme’s primary view is a Gantt chart whose rows are tracks. CPM [32] and PERT [41] added precedence networks, earliest and latest start times and three-point duration estimates. Rhylthyme’s variable durations are a three-point estimate without PERT’s distributional assumptions, and its `analyze_schedule` tool reports the dependency chain that determines the makespan rather than CPM’s float-based criticality (Section 5).

Resource-constrained project scheduling. The resource-constrained project scheduling problem (RCPSP) generalizes CPM with renewable resources of bounded capacity [12, 34, 54]; in its terms a Rhylthyme track is a capacity-one resource and each resource constraint is a renewable resource of capacity `maxConcurrent`. RCPSP solvers optimize makespan or cost, whereas Rhylthyme checks feasibility, surfaces conflict windows and applies staggering heuristics, leaving the final timing to the person running it. That division of labour follows the mixed-initiative and interactive scheduling tradition [28, 49, 79], but not its reactive-rescheduling half [31]: at run time Rhylthyme only propagates delays along dependency chains (Section 5).

Real-time flow management in air traffic control. Air-traffic flow management is the archetype of scheduling that is both resource-constrained and continuously replanned: an NP-hard integer program over airport and sector capacities [10], operated under Collaborative Decision Making (CDM) [8, 73]. Two ideas transfer at a much smaller scale: capacity as a cap on concurrent occupancy of a shared resource, which is Rhylthyme’s `maxConcurrent`, and CDM’s premise that the plan is a shared artifact updated as participants act, which is the premise of Rhylthyme’s shareable timeline. Rhylthyme has no optimization layer and does not yet merge progress across participants.

Temporal reasoning and robot task scheduling. Rhyllhyme’s triggers are temporal constraints between intervals: Allen’s algebra names the qualitative relations [2], OWL-Time gives them the web vocabulary the schema is annotated with [27, 72] (Section 4), and a program with fixed durations, non-negative offsets and conjunctive triggers is a simple temporal network [19] whose consistency the validator checks by resolving all start times. Temporal networks with uncertainty model durations the executor does not control [47, 70], whereas Rhyllhyme’s variable and indefinite steps are durations the executor *does* control (Section 9). Multi-robot task allocation and human–robot team scheduling [22, 23, 36] solve the assignment problem that Rhyllhyme’s environments only declare and account for, and behavior trees [14] are the reactive-execution counterpart of its runner, as control-flow trees rather than timed parallel tracks.

Domain schedulers: kitchens and laboratories. Multi-dish cooking has been scheduled since branch-and-bound food-service planning [25], and systems from Cooking Navi onward turn recipes into task graphs and present one merged procedure to the cook [26, 42, 48, 64]; each computes a procedure and presents it once, whereas Rhyllhyme keeps the resource model live during execution, with a clock, cues, manual steps and re-runs. In the laboratory, self-driving labs have made scheduling a first-class concern [17, 21, 38, 55, 63, 81], and recent work translates protocols into dependence graphs [58] or lets an LLM agent author and validate them [5], in both cases for robotic execution. Rhyllhyme sits at the human end of a stack that runs from device drivers [52, 59, 75] through machine-readable protocol languages [4, 60, 61] to human-readable protocols and technician-in-the-loop execution [62, 71]: it imports from both ends and produces timed, resource-aware programs a scientist runs by hand, its zero-offset `afterStep` being the human-executed analogue of the SDL synchronization constraint.

Workflow engines and process notations. Configuration-based bioinformatics pipeline frameworks are the closest *architectural* relatives (a declarative description of steps and dependencies, a validator, an execution engine and importers); on the axes of our earlier survey [39] Rhyllhyme is an explicit-syntax, configuration-paradigm system with both a command line and a web workbench. What separates it from them, from an MCP-native engine that executes an agent-written JSON blueprint of tool calls [53], and from McKeever et al.’s platform-agnostic JSON workflow scheduler with its Nextflow importer and graphical front end [43] is the executor: those run on machines and optimize for reproducible output, while a Rhyllhyme program runs on a person and optimizes for a timeline that can be followed. BPMN 2.0 [50] is the nearest human-process notation, with user tasks and timer events, but it leaves capacity limits on shared resources and wall-clock playback to the engine that hosts it.

Language models as authors of procedures. Closest to Rhyllhyme’s agent interface is work in which a language model turns procedural text into a structured artifact that can be checked. Kourani et al. have sixteen models write business-process models from text in an intermediate language, feed validator errors back to the model, and find that efficient error handling goes with model quality [37]; PAGED extracts procedural graphs from 3,394 documents and finds that models identify actors and actions well but can hardly pass 0.5 F1 on the gateways and flows between them [20], the same gap between finding steps and relating them that Rhyllhyme’s structured authoring prompt is built to close (Section 7). BioPlanner tests whether a model can reconstruct a biology protocol as pseudocode [51], and NATURAL PLAN shows that frontier models solve about a third of trip-planning problems stated in prose and almost none at ten cities [80]; none of these combines timed steps, capacity limits on shared resources and a human executor.

3 The Rhyllhyme program model

A Rhyllhyme program is a JSON document validated against a JSON Schema (version 0.2.0-alpha at the time of writing). Figure 1 shows a five-track Thanksgiving dinner planned around one oven, rendered by the bundled timeline renderer; Listing 1 shows three of its steps. Table 1 defines the concepts the language shares with every schedule of this kind, with one example from each vertical.

Listing 1: Three steps from Figure 1: an indefinite roast that occupies the oven until the cook ends it; a step that starts 45 minutes before the roast is due to end; and a manual gate that the host fires when guests sit down.

```
{ "stepId": "turkey-roast", "name": "Roast until 74°C", "task": "oven",
  "duration": { "type": "indefinite", "defaultSeconds": 9900 },
  "startTrigger": { "type": "afterStep", "stepId": "turkey-prep" } },
```

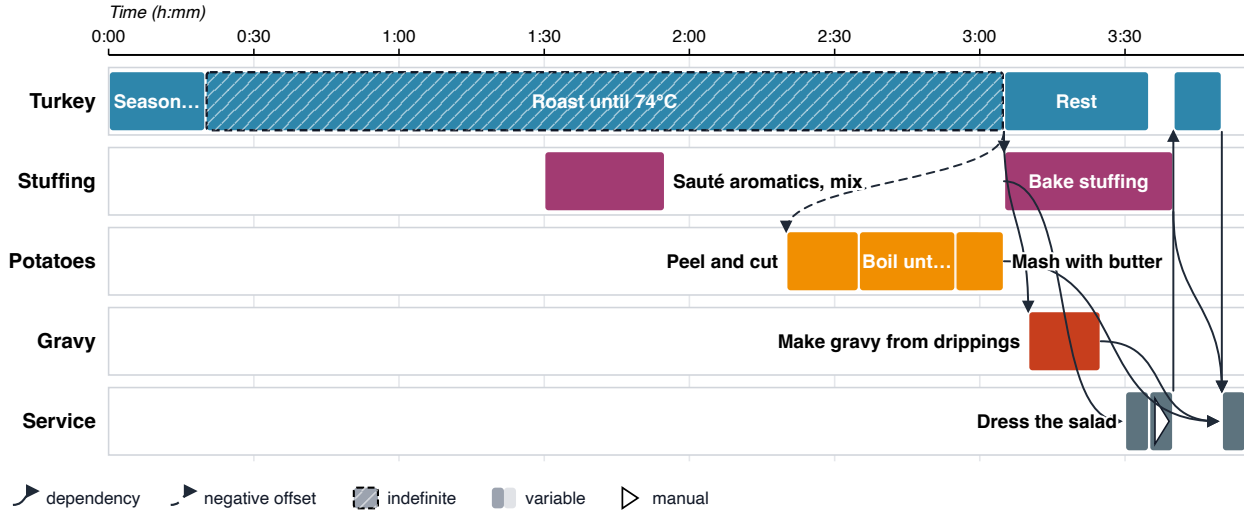


Figure 1: *Thanksgiving with One Oven* (five tracks, thirteen steps, 3 h 55 at planned durations) as rendered by the open-source timeline renderer (@rhylthyme/timeline) that also produces the MCP server’s image previews. Rows are tracks; bars are steps positioned by resolved start time. The roast is *indefinite* (hatched, dashed border): it ends when the cook says so, and everything downstream is planned against its default duration. Arrows are cross-track dependencies; the dashed arrow is a negative offset (peel the potatoes 45 min before the roast is due out). The flag on “Guests seated” marks a *manual* gate that the host fires; “Carve” waits for it, and “Serve” waits for all four dishes. The oven is a capacity-one resource, so the stuffing bakes only after the turkey comes out. The web application draws its own D3 timeline from the same JSON.

```
{ "stepId": "potatoes-peel", "name": "Peel and cut", "task": "prep",
  "duration": { "type": "fixed", "seconds": 900 },
  "startTrigger": { "type": "afterStep", "stepId": "turkey-roast",
    "offsetSeconds": -2700 } },
{ "stepId": "guests-seated", "name": "Guests seated", "task": "host",
  "duration": { "type": "fixed", "seconds": 300 },
  "startTrigger": { "type": "manual", "triggerName": "guests-seated" } }
```

Tracks, steps and triggers. A program has a `programId`, `name`, optional `environmentType`, optional `actors` or an `environment` reference, an array of `tracks`, optional `resourceConstraints` and free-form `metadata` (ingredients, serves, source attribution, thumbnail). A track is a named sequence of steps that must not overlap in time; the array order of steps is not itself enforced, only non-overlap. Step identifiers are unique across the whole program so that any step may reference any other. A start trigger is one of `programStart`, `programStartOffset`, `afterStep` (anchored to the referenced step’s end, or its start with `event: start`, plus a signed `offsetSeconds`), `afterStepWithBuffer`, `manual` and `onAbort`, or a compound `{logic: all|any, triggers: [...]}` that waits for all, or the first, of several. Offsets and durations accept a number of seconds or a unit string (“5m”, “1h30m”). A trigger may carry a `choiceId`; a step with a `choice` block presents options at run time, and downstream steps gated on a `choiceId` run only for that option. A negative `afterStep` offset (“start 20 minutes before the roast is done”) is accepted only against an indefinite step; planners place it against the referenced step’s `defaultSeconds`, and the runtimes fire it from that forecast rather than from the actual end, a limitation discussed in Section 9.

Durations. A *fixed* duration has `seconds`. A *variable* duration has `minSeconds` and `maxSeconds`, an optional `defaultSeconds` used for planning, and may be ended early by the executor. An *indefinite* duration runs until the executor ends it; `defaultSeconds` gives planners something to draw. In every case the person running the program decides when a non-fixed step ends; the language has no construct for a duration that is uncertain *and* outside the executor’s control.

Listing 2: A home kitchen with two cooks, trimmed to three of its sixteen resource constraints: the oven is capacity-one and takes half a cook’s attention, the four burners take a whole cook each, and the microwave a quarter.

```
{ "environmentId": "home-kitchen", "name": "Home Kitchen",
  "type": "kitchen", "actors": 2,
  "actorTypes": {
    "home-cook": { "name": "Home Cook", "count": 2 } },
  "resourceConstraints": [
    { "task": "oven", "maxConcurrent": 1, "actorsRequired": 0.5,
      "qualifiedActorTypes": ["home-cook"],
      "description": "Single oven (requires partial attention)" },
    { "task": "stove-burner", "maxConcurrent": 4, "actorsRequired": 1.0,
      "qualifiedActorTypes": ["home-cook"],
      "description": "4-burner stovetop (one cook monitors all four)" },
    { "task": "microwave", "maxConcurrent": 1, "actorsRequired": 0.25,
      "qualifiedActorTypes": ["home-cook"],
      "description": "Microwave oven (minimal attention needed)" } ] }
```

Environments. An environment is a separate JSON document, validated against its own schema, that describes a workplace once so that many programs can share it. It has an `environmentId`, a `type` from a controlled list (37 values, from `kitchen` and `laboratory` to `theater`, `gym` and `datacenter`), `actorTypes` (named categories of people with a `count`), and `resourceConstraints`. Each constraint names a task, its `maxConcurrent` capacity, the fraction of a person it occupies (`actorsRequired`: an oven needs 0.5 of a cook, a microwave 0.25, a stove burner 1.0), and which actor types are `qualifiedActorTypes` for it. A program references an environment by id; the loader falls back to the default environment for the program’s `environmentType` when none is named, and inline `resourceConstraints` on the program take precedence. The repository ships ten environments (home and commercial kitchens, academic, biotech and pharma laboratories, a bakery, an airport, a software team), and `create_environment` on the MCP server produces new ones. A program that names neither an environment nor `actors` must declare a resource constraint for every task it uses. Environments are what let the same protocol be scheduled differently in a lab with one thermocycler and a lab with four; the attention and qualification fields are honoured by the command-line runner, while the web and MCP engines use only capacities and the program-level actor count.

Listings 2 and 3 show two of them, trimmed to a few constraints each. A program picks an environment up in one of two ways: it names one in an `environment` field, or it declares an `environmentType` and the loader resolves that to the type’s default environment. The consequence is that the same program runs unchanged in different workplaces — the breakfast program of Section 6 against the home kitchen of Listing 2 or against the six-burner commercial kitchen — and what changes is capacity, not procedure: a schedule that is comfortable in one may be over-utilized in the other. The `actorsRequired` fractions are where attention rather than equipment is booked. A stove burner takes a whole cook, an oven half of one and a microwave a quarter, so the two cooks of Listing 2 can hold two burners, or a burner, the oven and the microwave, but not two burners and the oven; Listing 3 books a fume hood the same way (an entire person, against less than a third of one for a thermal cycler that runs itself) and restricts the hood and the centrifuge to the actor types trained for them, which the undergraduate is not. These fractions are the attention model referred to in Section 8.

Replicates and staggered starts. Repetition has a shorthand so that authors need not copy tracks. A track may declare `batch_size` and `stagger`: `"batch_size": 3, "stagger": "1m30s"` expands into three copies of the track whose starts are 90 seconds apart, and `templateId` with `templateParameters` lets those copies instantiate a shared `trackTemplates` entry with different values (three landings on one runway with different flight numbers, in the airport example). A track or step may instead declare `replicates` with a `count`, a `mode` of `parallel`, `stagger` or `serial`, and a `delay`; step-level parallel replicates fan out into sub-tracks that rejoin, serial replicates chain within the track, and staggered replicates delay each start by the given amount. Expansion happens before validation in both toolchains, so the validators, planners and runtimes only ever see plain tracks and steps.

Schema 0.3.0-alpha adds three constructs that say what happens *downstream* of a replicated step, per instance. A step-referencing trigger may carry `instances`: `"each"` replicates the dependent step once per instance and

Listing 3: A small academic laboratory, trimmed to three of its twenty-seven constraints and three of its seven actor types. Two thermal cyclers run almost unattended; the fume hood (safety training required) and the centrifuge are restricted to qualified actor types, which exclude the undergraduate.

```
{ "environmentId": "small-academic-lab",
  "name": "Small Academic Laboratory", "type": "laboratory",
  "actorTypes": {
    "postdoc": { "name": "Postdoctoral Researcher", "count": 1 },
    "grad-student": { "name": "Graduate Student", "count": 2 },
    "undergrad": { "name": "Undergraduate Student", "count": 1 } },
  "resourceConstraints": [
    { "task": "pcr-thermal-cycler", "maxConcurrent": 2, "actorsRequired": 0.3,
      "qualifiedActorTypes": ["postdoc", "grad-student", "undergrad"],
      "description": "PCR thermal cyclers (96 wells, simple operation)" },
    { "task": "fume-hood", "maxConcurrent": 1, "actorsRequired": 1.0,
      "qualifiedActorTypes": ["postdoc", "grad-student"],
      "description": "Chemical fume hood (4 ft, safety training required)" },
    { "task": "centrifuge", "maxConcurrent": 1, "actorsRequired": 0.4,
      "qualifiedActorTypes": ["postdoc", "grad-student"],
      "description": "High-speed centrifuge (24 x 1.5 mL capacity)" } ] }
```

pairs instance i with instance i of its predecessor (each tray of cookies starts cooling the moment it leaves the oven), "all" is a barrier that waits for every instance, and "any" fires on the first — the multiple-instance, synchronization and discriminator patterns of the workflow-pattern catalogue [68], written on a trigger rather than drawn in a graph. The default is "all", which is the rejoin that earlier schema versions already performed, so existing programs keep their timings. A replicated step may also declare **maxInFlight**: a bound on how many instances may be *in flight* between it and its barrier, where an instance is in flight from its own start until it has finished every "each" descendant. This is not **maxConcurrent**, which bounds occupancy of one task at one instant; **maxInFlight** bounds a chain of tasks across time, and the difference is operational rather than cosmetic. A cooling rack that holds two trays, declared as **maxConcurrent**: 2, would let a third tray bake and then leave it hot with nowhere to go; declared as **maxInFlight**: 2 it holds the third *bake* until a tray leaves the rack. Hold the upstream step, do not strand the downstream one. The same shape is a centrifuge rotor that holds six tubes, a bench with room for four open plates, or a taxiway between landing and gate. The three constructs are McKee et al.'s asynchronous nodes, barrier nodes and maximum asynchronous concurrency [43], carried over from compute workflows, where the bounded resource is disk space, to human-executed schedules, where it is oven racks, rotors and bench space. Expansion rewrites all three into plain **afterStep** and **compound** triggers, the same reduction of higher-level routing constructs to a small primitive set on which Petri-net workflow semantics rest [67] — the in-flight bound becomes a synthetic conjunction on instance $i + k$ that also waits for the last descendant of instance i — so the addition costs the validators, planners and runtimes nothing beyond the expander pass, and because **afterStep** fires on actual completion, a tray that cools slowly really does hold up the next bake.

4 Ontologies and vocabularies

The program and environment schemas carry **\$comment** annotations written in the vocabulary of the W3C Time Ontology in OWL [27, 72], itself grounded in Allen's interval relations [2]. A step is a **time:ProperInterval** with a **time:hasBeginning** and **time:hasEnd**; a fixed duration is a **time:Duration** with **time:numericDuration** in **time:unitSecond**; **afterStep** with zero offset is **time:intervalMetBy** and with a positive offset **time:intervalAfter**; a pre-buffer **time:intervalMeets** its step and a post-buffer is **time:intervalMetBy** it; **programStartOffset** places a **time:Instant** relative to program start; compound triggers are conjunctions or disjunctions of these relations. The annotations are documentation, not an RDF export: they fix the intended meaning of each construct for implementers of the validator, the two runtimes and the renderer, and they make the mapping to the temporal-reasoning literature explicit rather than folkloric. Generating OWL-Time individuals from a program is a small transformation we have not built. A second mapping, to provenance rather than time, is just as direct: a step is a PROV-DM Activity, a resource an Entity and an actor an Agent [45], which is the vocabulary in which a run record would be exported. A more ambitious use of the same annotations

would compile them into the tool interface itself, so that an agent could only construct programs that satisfy the ontology’s constraints; Zhou et al. demonstrate that ontology-to-tools compilation for LLM agents on a knowledge-graph platform [82]. Rhyllhyme enforces its constraints after generation, through the validator, rather than during it.

Three smaller controlled vocabularies do practical work. The environment **type** list names the kinds of workplace the system knows how to default. An instrument catalog groups laboratory and kitchen equipment into categories (centrifuges, thermal cyclers, plate readers, liquid handling, cold storage, ovens, ranges, mixers, proofers, and so on) with default capacities and attention requirements, so that an environment can be assembled by picking instruments. And the public recipe catalog — an imported corpus far smaller than the recipe datasets assembled for text generation [11] — is faceted by a fixed taxonomy of categories, cuisines, diets, time buckets, main ingredients, methods and tags, which the search API validates strictly so that a mistyped facet fails loudly rather than silently returning nothing.

5 Runtime and tooling

Validation. There are two validators. The Python one, used by the command-line tools, runs JSON Schema validation and then logic checks: duplicate or missing identifiers, references to non-existent steps, choice references to options that do not exist, tasks without a resource constraint, and, after resolving every start time, steps that overlap within a track. The JavaScript one, used by the MCP server, performs the same logic checks without the schema pass, adds dependency-cycle and unparseable-duration detection, and reports each finding with a stable code, a message and a *fix* hint (“chain **s2** with {**type**: **afterStep**, **stepId**: **s1**} or move it to its own track”), which matters when the author is a language model that will act on the message verbatim. The two share one timing semantics: a parity test resolves every program in the example corpus with both and asserts identical start and end times, currently 616 steps across 39 programs. The top-level corpus holds 38 JSON programs (15 kitchen, 6 laboratory, 4 event, 3 fitness, 2 bakery, 2 airport, 1 manufacturing, 5 general or test); both validators reject the same three, two for within-track overlaps and one for tasks with no resource constraint.

Common optimizations. The auto-planner, available from the web editor and as a Python module, works from earliest start times under the dependency graph, checks the result for resource over-utilization (intervals where more steps occupy a task than its capacity allows), and offers four named strategies that correspond to what people actually ask a schedule to do: *synchronized finish* delays shorter tracks so that every track ends at the same moment (dinner on the table together); *minimize length* packs steps as early as dependencies allow; *provide consistent activity coverage* sequences steps one after another, alternating tracks, so that a single executor always has something to do and no long idle gaps open up; and *fit to time* compresses flexible steps and omits low-priority ones until the program meets a target duration, using each step’s **priority** and **flex** settings (a flexible step may **fill** the gap to its successor or stretch to its **maxDuration**). All four are heuristics over earliest-start times and dependency order, not solvers; they report over-utilization rather than resolving it, and their job is to hand the person a feasible schedule and to say where the pressure is. (An older planner in the core library predates the current schema and is not described here.)

Analysis. **analyze_schedule** resolves all start times and computes the makespan; the chain of predecessors that determines it (walking back from the last-ending step through the predecessor whose end is closest to each start); resource-conflict windows by a sweep line over each task’s occupancy; peak concurrency against declared actors; per-track slack before the program’s end; and, given a **finishAt** or **startAt** instant, the wall-clock start of every step. “When do I put the potatoes in so that everything is ready at six” is a single call. Each conflict window is tagged with the **kind** of limit it violates, **maxConcurrent** or **inFlight**, and a **bindingConstraints** list names, for every edge of the critical chain, which limit actually gates it: for the three-tray example above the analyzer reports the rack rather than the oven, which is the answer the cook needs.

Execution. Two runtimes play a program. The command-line runner is a curses interface that advances a clock, starts steps as their triggers fire, waits on manual triggers, lets the executor end variable and indefinite steps, accounts for actor capacity from the environment, and handles abort with **onAbort** triggers. Because **afterStep** triggers fire on actual completion, a step that runs long delays everything chained after it, while steps anchored to program start by offset stay on the clock; there is no replanning beyond that propagation. The web

application is a player: it advances a simulated clock on a browser interval timer at a chosen speed, starts and highlights steps as their resolved times arrive, and lets the executor end non-fixed steps; it is not re-anchored to wall-clock time while running, so a phone that sleeps mid-roast pauses the schedule rather than catching up. It renders the program in three principal views and several auxiliary ones (Section 6). Both runtimes consume the same JSON, and both are what the MCP server links to.

Importers. A registry of importers, plus a separately authenticated Benchling connector, converts external sources into programs: Spoonacular and TheMealDB recipes, Cooklang files, protocols.io and Benchling protocols, and Opentrons Protocol API scripts. Importers infer tracks and durations from source structure, a much weaker form of the protocol-to-dependence-graph translation that Shi et al. pursue for robotic labs [58] or of the flow-graph parsing that the language-processing community has built for recipe text [33, 46], and that bounds their quality: a recipe or protocol source typically yields one or two tracks, and the multi-track programs in the corpus are authored by hand or by an agent. A language model is the stronger extractor: Almunashiri et al. show that GPT-4o can infer a structured provenance record of an experiment from the text of a biomedical paper [3], the same move from prose to a machine-usable process description that an agent makes when it reads a recipe or protocol and authors a program through the MCP server (Section 7). The output is validated like any other program. The Opentrons importer runs the protocol script in the Opentrons simulator to recover its command sequence, which executes untrusted code and is sandboxed only by process isolation on the server; we return to this in Section 9.

6 Views in the web application

Timeline. The timeline is a Gantt chart with one row per track and one bar per step, positioned by resolved start time and coloured by track (Figure 2; Figure 1 is the static form). It carries the execution controls: start, pause and stop, a speed multiplier for rehearsal, a zoom slider, and a time-format selector that switches the axis between elapsed minutes and time of day from a chosen start. While running, a cursor sweeps the chart, active steps are highlighted, audio cues fire at step boundaries, and variable and indefinite steps expose a control to end them. On the kitchen vertical an ingredient checklist sits beside it.

Itinerary. The itinerary is the same schedule sorted by clock time regardless of track: a chronological list of “at 17:12 start the potatoes” entries with each step’s track, duration and description. It is the view a person reads when they want to know what to do next rather than how the tracks relate, and it is the text form the MCP server returns in its previews. Figure 3 (left) shows it for the program of Figure 2.

Dependency graph. The DAG view draws steps as nodes and triggers as edges, with buffered edges styled distinctly, so that cross-track dependencies and choice branches, which the timeline can only show by position, are explicit. It is the view used when editing a program’s structure. Figure 3 (right) shows it for the breakfast program, and Figure 4 for a laboratory one.

Others. A resources view plots each task’s concurrent usage against its capacity over time, which is where over-utilization is seen; a radial view wraps the timeline around a clock face; a kanban view groups steps by state during execution; a fold view collapses the program into the sequence of merge points; an editor view exposes drag-to-reschedule and the auto-planner strategies of Section 5; and a video view plays step media where a program provides it. The web application also hosts a public catalog of programs, per-user libraries, and per-vertical hosts that differ only in defaults and copy.

7 The MCP server: agent-native authoring

The Model Context Protocol [6] standardizes how a language-model host discovers and calls tools, reads resources and invokes prompts; the server described here uses the Streamable HTTP transport and tool annotations introduced in the 2025-03-26 revision and the structured tool output of 2025-06-18 [44]. It is deployed as a stateless endpoint and served at five paths, `/mcp`, `/kitchen/mcp`, `/lab/mcp`, `/events/mcp` and `/gym/mcp`, from one implementation: each vertical differs from the generic endpoint only in default catalog filter, viewer host,

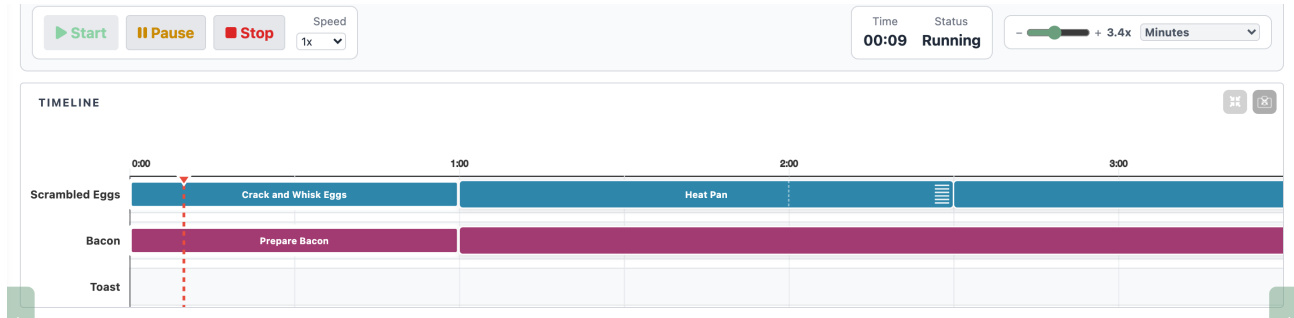


Figure 2: The timeline view of *Breakfast Schedule* (three tracks) nine seconds into a run. The control bar carries start, pause and stop, a speed selector (here 1×), the elapsed time and run status (*Running*), a zoom slider and a time-format selector that switches the axis between elapsed minutes and time of day. The dashed red cursor sweeps the chart and stands at 0:09, over the two steps that are active (“Crack and Whisk Eggs” and “Prepare Bacon”). “Heat Pan” is a variable step (60–120 s, planned at 90): the dashed line inside its bar marks the minimum, and the hatched segment at its end marks the part the executor may cut short. The Toast row is still empty because “Make Toast” starts 480 s after program start.

copy, and two added convenience tools (a one-shot “find the best match and open it” verb such as `cook_recipe` or `run_protocol`, and a random pick). Figure 5 shows the intended interaction.

Tools. Each of the thirteen tools is registered with a title and MCP annotations (`readOnlyHint`, `destructiveHint`, `idempotentHint`, `openWorldHint`); four (validation, analysis, search and publication) also carry an output schema and return `structuredContent`, so that hosts which honour annotations can auto-approve read-only calls and agent frameworks can consume results without parsing markdown. The annotations are self-declared hints, and a host is free to ignore them. `validate_program` and `analyze_schedule` are pure and read-only. `visualize_schedule` validates first and by default refuses an invalid program with the same coded findings (an explicit `allowInvalid` flag publishes a rough draft anyway), then creates a share record and returns a markdown preview (cover image, equipment, ingredients, an ASCII Gantt, the itinerary and a schedule check), an inline PNG of the timeline, and the live URL. `import_from_source` wraps the importer registry; `create_environment` produces an environment document; `search_public_recipes` and `load_public_recipe` read the public catalog; `login`, `list_my_programs`, `load_program` and `save_program` handle a user’s own library, with `save_program` also validating first. Failures set `isError` and say what to do next, including when a call needs a login token.

Resources and prompts. The server publishes the program JSON Schema, a one-page authoring guide (the rules the validator enforces, the trigger vocabulary, how to make tracks finish together, choice branching), and five complete example programs as MCP resources, and a `plan_schedule` prompt that takes a goal, an optional deadline and optional resource limits and walks the model through the workflow in Figure 5. Server-level `instructions` carry the workflow once per session rather than repeating it in every tool description.

Why this matters. In our experience the failure mode of an LLM asked to “plan Thanksgiving with one oven” is a wall of prose that is wrong about timing and impossible to follow at the stove. Giving the model a validator with fix hints, an analyzer that answers “when do I start,” and a publisher that refuses malformed input by default moves the burden of *structural* correctness (identifiers, dependencies, overlaps, declared resources) from the model’s text to the tool contract, and the deliverable from a chat message to a page on the cook’s phone. Whether the durations and orderings the model chooses are sensible is not something the validator can check. This is the LLM-Modulo arrangement, in which the model generates candidates and external, sound critics verify them rather than the model being trusted to plan [30], a division the planning benchmarks motivate directly [66]. The pattern is not unique to Rhylythme: Benjamin et al.’s Planning Copilot exposes syntax checking, planner selection, plan validation and execution simulation to an LLM as MCP tools for the same reason [9], and Angelopoulos et al. pair natural-language protocol authoring with automated validation for a lab orchestrator [5]. What Rhylythme adds is the domain (a person’s timeline rather than a planner’s plan or a robot’s protocol) and the publication step. We now report it as a measurement. Following Almontashiri et

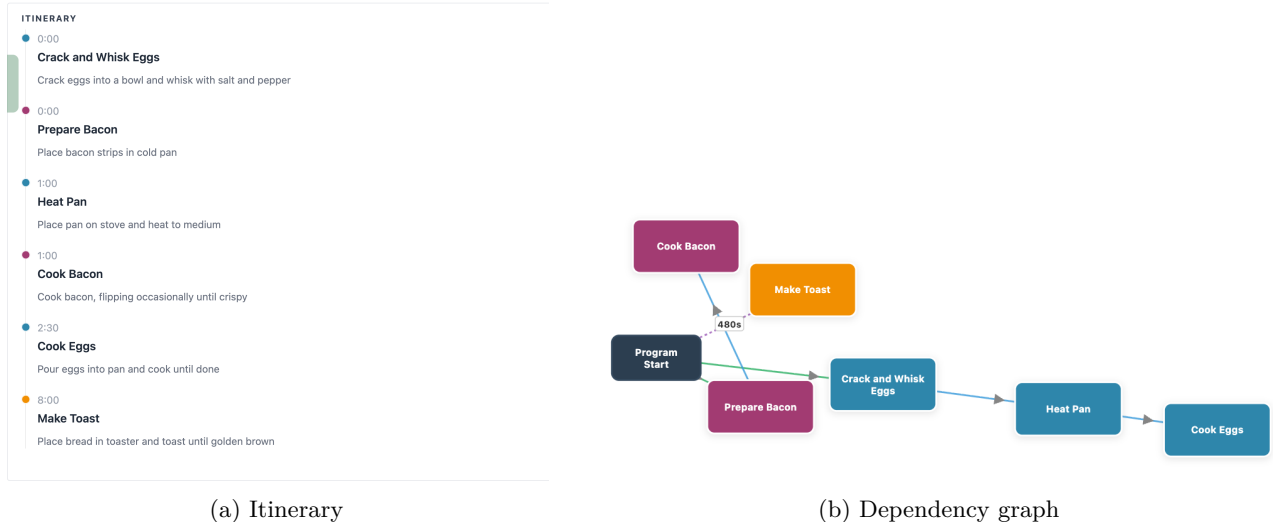


Figure 3: Two more views of the program in Figure 2. (a) The itinerary flattens the schedule into one chronological list of start time, step, track colour and description: the form a person reads at the stove or the bench to know what to do next. (b) The dependency graph shows the same program by structure rather than by clock: solid edges are start triggers from program start or from a preceding step, and the dashed edge labelled 480s is the `programStartOffset` of “Make Toast,” the one step pinned to the clock rather than to another step.

al.’s protocol for extracting structured records from text [3], and the question-refinement and cognitive-verifier prompt patterns catalogued by White et al. [74], we replaced the single `plan_schedule` message with four turns—read-back, target-model check, extraction of steps with the source span each came from, the schema-constrained, span-attributed extraction Dagdelen et al. apply to scientific text [16], and only then relationships—and scored both against 24 expert-authored programs (10 kitchen, 6 laboratory, 4 event, 4 fitness), each paired with its source text and per-step spans. On `claude-haiku-4-5` the four turns raise relationship (trigger) F1 from 0.22 to 0.63 and the end-to-end pass rate—both validators pass, makespan within 10% of the expert’s, critical chain sharing at least half its steps—from 0.08 to 0.62. The gain is in extraction rather than in inference: asking for the step list before any dependency exists lifts step recall from 0.37 to 0.95 and cuts the unsupported (unattributable) step rate from 0.60 to 0.01, so the triggers have real steps to point at; relationships nevertheless remain the weakest component, as the paper also found. The gold set, the harness (`rhylthyme eval-prompts`) and the committed per-program numbers ship with the repository, so a prompt change is gated on them in CI.

8 Discussion: positioning

Summarizing the comparison, *Rhylthyme* *borrow*s the Gantt row-per-track grammar, the RCPSP resource model, the idea of a critical chain, and the trigger vocabulary of temporal constraint networks. It *omits* what most of the compared systems are built around: it does not optimize an objective, does not reason about durations outside the executor’s control, does not solve actor assignment, and does not execute anything on a machine. What is *new* is narrower than a new model of human work: in the language and both runtimes the person is a capacity-one resource whose duration oracle is a button, and the human-specific constructs are the triggers that wait for that button, the attention fractions in environments, and the affordances of the runtimes. The claim is that no compared system combines capacity constraints and cross-track dependencies with a followable, human-executed timeline that has manual, executor-ended and abort steps; that the same schema is used unchanged across cooking, laboratory, event and training verticals, with domain content kept in metadata and environments rather than in the schema; and that validation, analysis and publication are exposed as typed, annotated tool calls rather than as instructions in a prompt. The comparison is with traditions rather than with individual systems: it describes the mainstream of each, and specific systems (a workflow platform’s MCP server, a lab orchestrator’s guided manual steps) already occupy some of the ground claimed here.

The nearest neighbours deserve a direct comparison. The cooking-step schedulers of Matsushima and Funabiki and of Nakabe et al. [42, 48] and the SDL scheduler of Zhou et al. [81] model the same objects (parallel procedures,

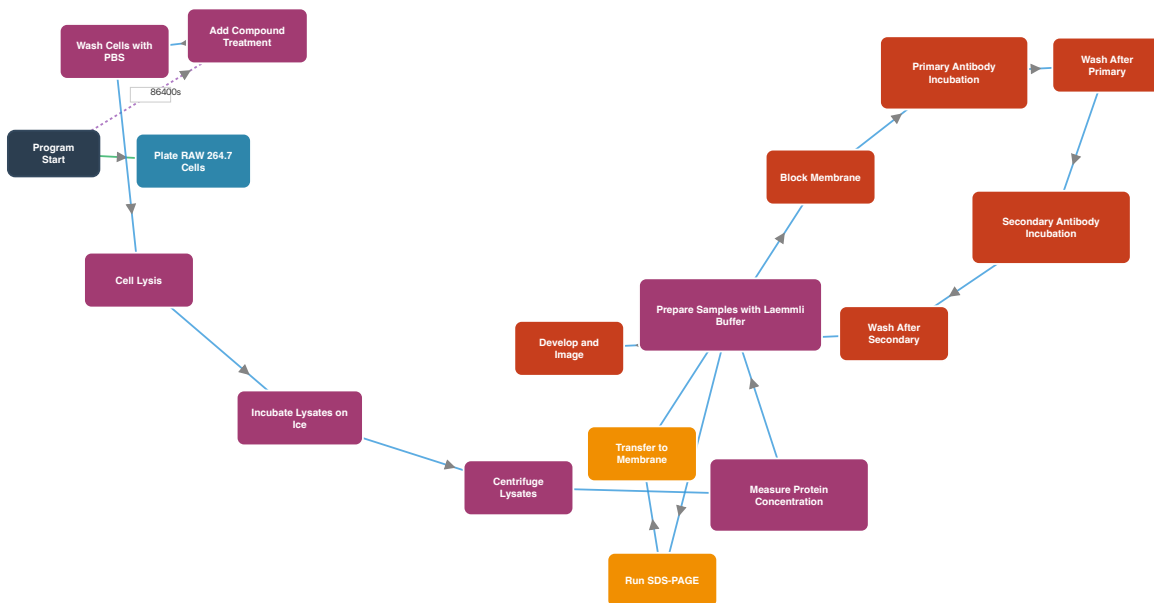


Figure 4: The same dependency-graph view on a laboratory program, *Protein Lysate and Immunoblotting* (four tracks, sixteen steps over two days). Read left to right it follows the bench: cell preparation (blue) and treatment and lysis (magenta) on the left, gel electrophoresis (orange) at the bottom, and the immunoblotting chain of blocking, primary and secondary antibodies and washes (red) on the right. The dashed edge labelled 86400s is the `programStartOffset` that holds “Add Compound Treatment” until day 1. Nothing in the view is specific to kitchens: like the schema, it is vertical-independent. Exported from the web application’s D3 view as vector graphics.

shared stations, tight synchronization) and solve for an optimized procedure; Rhyllhyme would be a natural *target* for their output, since none produces a shareable runtime or an agent interface. Protocol run modes and guided-cooking timers are the other neighbour: they have the runtime but not the model, so a second dish or a second protocol cannot be placed against the first. The mixed-initiative scheduling workbenches [28, 49] had both a model and a human in the loop, but the human’s role was to edit the schedule, not to execute it. Workflow engines and lab orchestrators, finally, would be natural *consumers* of Rhyllhyme’s trigger vocabulary if the executor were a machine; the language is agnostic about that, but the runtimes are not.

9 Limitations and future work

The auto-planner is a heuristic with no optimality guarantees. Variable and indefinite steps are ended by the executor, so a program’s uncertainty is under the executor’s control; the case the temporal-networks literature treats, a step whose end is decided by a process (an incubation, a rise), is not represented, and adding it would make dynamic-controllability checking [47] the right tool for “start twenty minutes before the roast is done.” Today that construct is only partly honest: a negative offset has to fire before its anchor ends, so both runtimes resolve it against a *projection* of that end rather than the end itself. The projection is the author’s `defaultSeconds` unless the program opts in with `metadata.offsetsUse` set to `predicted`, in which case an indefinite anchor is projected from the program’s own run history whenever a prediction exists whose interval is narrower than the authored default; an anchor that ends before the projected instant releases the trigger immediately, so a projection can delay work but never hold it back once the thing it was projecting has happened. A roast that finishes early or late is therefore compensated only to the extent that history predicts it in advance, and not at all once the run is under way. The two validators agree on timing but not on coverage: only the Python one applies the JSON Schema, and only the JavaScript one detects dependency cycles and unparseable durations; a single specification both implement is overdue. The web application’s clock is a simulated player clock, not a wall-clock dispatcher. Actor constraints are accounted for but not solved; a small assignment layer

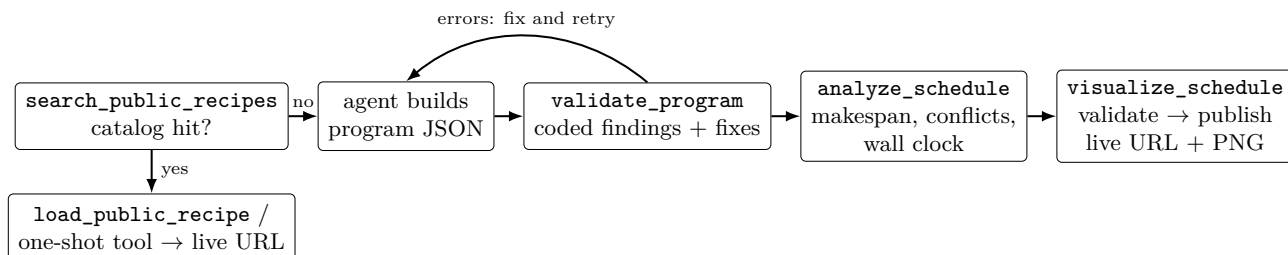


Figure 5: The workflow the server’s **instructions** describe to the host at initialization. Catalog content is returned with its live URL directly; new content is validated before it can be published.

along the lines of Tercio [23] is the obvious next step for multi-person kitchens and labs. When several people follow one program, each runs their own copy of the clock; merging progress across participants, the CDM idea, is not implemented, and the reactive rescheduling that the scheduling literature has treated as central since the 1990s [31] is limited to delay propagation. The rhythm of cues and manual gates is fixed by the program; the evidence that interaction timing shapes users’ agency [79] suggests it should be a design parameter. The reported critical chain is a heuristic and not a float-based critical path, and the web application’s live page computes step timings with its own routine, which the parity test does not yet cover. Import quality is bounded by source structure, recipe text under-specifies durations, and the Opentrons importer executes fetched protocol code, which needs a proper sandbox before it should accept arbitrary URLs. Durations are no longer only author guesses, which matters because authored estimates are systematically optimistic: people underestimate how long their own tasks will take, and do so even when they have the relevant history [13, 29]. Learning durations from recorded executions instead is what the process-mining literature does for remaining flow time, from annotated transition systems over event logs to stochastic Petri nets with arbitrary firing delays [56, 69]. Badosa et al. predict each bioinformatics job’s performance from its execution history under different combinations of cluster resources, and co-schedule jobs on one node only when they are compatible [7]. Both Rhythyme runtimes now write one run record per execution — the planned and the actual start and end of every step, what ended each non-fixed step, and the author-declared variance factors the executor answered at run start — and the same identical-then-similar lookup runs over those records: an identical context yields a median and an interval, otherwise a per-step least-squares fit on the factors whose correlation with the observed duration clears a threshold. A held-out evaluation, which trains on a program’s earlier runs and predicts its latest fifth, measures the result against the author’s number. On a synthetic corpus with a known generating law it recovers that law: for an indefinite step whose duration is a function of a declared factor the mean absolute error falls from 2 h 26 min, the authored `defaultSeconds`, to 81 s, a 99 % reduction; for a step whose observed durations were generated around the authored value itself, prediction is 26 % *worse*, because there the author’s number is already the true mean and history can only add estimation noise to it. Which of the two a real step resembles is an empirical question per step, decided by the same variance analysis Badosa et al. use to establish that a history is inferential at all; the corresponding measurement on real catalog runs awaits a thirty-day observation window and is not reported here. What remains absent is the reactive half: prediction changes when a negative offset fires and which durations the planners are shown, but nothing is rescheduled while a run is in progress. Compatibility-based co-scheduling is the machine counterpart of the fractional **actorsRequired** model and is likewise not implemented. The MCP server’s login is a pasted, short-lived bearer token that passes through the model’s context, outside the protocol’s OAuth framework; adopting it would remove that step and that exposure for consumer hosts. Three example programs fail both validators (two for within-track overlaps, one for undeclared resources), which is evidence both that the checks are useful and that authoring remains error-prone. Finally, the language has no formal semantics beyond the schema, its OWL-Time annotations and the reference validators; writing one down would make the controllability and assignment work well-posed.

Activity recognition: from timer to conductor. Every runtime described here learns what the executor is doing only from what the executor tells it: a tap on a manual gate, a “done” on an indefinite step. The complementary technology is activity recognition, which infers from video, audio or wearable sensors which action a person is performing [35]. It is furthest along in the operating room, where surgical phase and tool recognition from laparoscopic video is a mature line of work [65] and the surgical data science agenda explicitly

asks for workflow-aware assistance that knows where in the procedure the team is [40]; egocentric and multi-view datasets of kitchen and assembly work now provide the fine-grained action labels, procedural structure and mistake annotations that step-level recognition needs [18, 24, 57]. The bench laboratory is being treated the same way. ProBio is a multimodal dataset of molecular-biology lab work whose annotations are guided by the written protocol, with benchmarks for tracking transparent solutions and for action recognition that fuses video with the protocol text [15]; FineBio records 32 participants performing mock experiments and annotates them as a hierarchy of protocol, steps and atomic operations, observing that biology requires strict adherence to the step order while leaving the order of operations within a step free, which is exactly the track-and-step structure of a Rhythyme program [78]; and a step-aware safety monitor built on protocol-annotated lab video first aligns the footage to the protocol to find the current step, then checks the step-specific constraints [77]. Coupling a recognizer to a Rhythyme program is a natural next step, and the coupling runs both ways. A recognizer that emits step-start and step-end events (the tube is capped, closure has begun, the actor has left the stage) would end indefinite steps and fire negative offsets from the observed end rather than from any projection, authored or predicted, removing the limitation noted above, and would make actor availability observable rather than declared. In the other direction the program is a strong prior: at any moment only a few steps can legitimately start, so the label space the recognizer must discriminate collapses from every action in the vocabulary to the handful the schedule allows, the ordering constraint that surgical phase models exploit and the reason ProBio and its successors condition recognition on the protocol. With both halves in place Rhythyme would act as a conductor for a team rather than a timer for an individual: it would see every track at once, cue each technician, nurse or stagehand at the moment their step becomes ready, and replan the remaining tracks when someone runs long, in a bench laboratory with several people on one protocol, an operating room, or a theatre where cues follow the action on stage rather than a stopwatch. The obvious risks are that a recognizer which advances the clock wrongly is worse than a timer, and that cameras in these rooms raise privacy and consent questions; the mixed-initiative reading is that recognized transitions should be proposed and confirmed until their reliability is measured, which the runtimes’ manual gates already support. None of this is implemented; it is the direction in which the `manual` and `indefinite` constructs, and the actor model, were designed to be extended.

10 Conclusion

Rhythyme is a small, opinionated answer to a common situation: a person, a clock, several things happening at once, and not enough ovens. It reuses well-understood ideas from project scheduling and temporal reasoning, packages them in a schema that survives being moved between kitchens, laboratories, stages and gyms, plays the result live with the resource model still in force, and, through MCP, lets a language model do the authoring against a tool contract that refuses by default to publish what will not work.

Availability. The specification, example programs and environments, the command-line runner, the timeline engine (`rhylthyme-timeline`) and the MCP server implementation (`rhylthyme-mcp`) are public at <https://github.com/rhylthyme>; the web application and importers are being prepared for release under the same organization. The remote server is at <https://mcp.rhylthyme.com/mcp>. This report describes the repositories as of September 2026 and is released under CC BY 4.0.

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Concept	Definition (Rhylthyme construct)	Examples
Program	The whole schedule: a named set of tracks plus its resources and metadata (programId , tracks , resourceConstraints)	Thanksgiving dinner; a Western blot; a wedding run-of-show; a 45-minute HIIT session
Track	One line of work whose steps run one after another; parallel work goes in separate tracks	The turkey; the gel; the ceremony; the lower-body superset
Step	A single timed activity within a track (stepId , duration , task)	Roast 3 h; incubate 1 h; procession 5 min; 12 reps
Dependency	A condition on when a step may start, within or across tracks (startTrigger : after a step ends or starts, at an offset from program start, on a manual signal, on abort, or all/any of several)	Gravy starts when the turkey comes out; blocking starts after transfer; music cue fires when the doors open; rest starts after the last set
Offset	A signed delay applied to a dependency; negative offsets start a step before its anchor ends	Potatoes in 20 min before the roast finishes; pre-warm the reader 10 min before the plate is ready
Buffer	Setup or cleanup time attached to a step, or an explicit gap between two steps (preBuffer , postBuffer , afterStepWithBuffer)	Preheat before baking; centrifuge spin-down; stage reset between acts; equipment changeover
Fixed / variable / indefinite duration	Exact time; bounded time the executor may end early; open-ended time the executor ends	Bake 25 min; sauté 3–5 min; incubate overnight until confluent; hold until the speaker finishes
Resource and capacity	A named task type with a limit on concurrent use (task , maxConcurrent)	One oven; two thermocyclers; one PA system; one squat rack
Actor and attention	People available to attend steps, by type and count, and the fraction of a person each resource needs (actorTypes , actorsRequired , qualifiedActorTypes)	A stove burner needs one cook; a microwave a quarter of one; only a licensed operator runs the cytometer
Environment	A reusable document declaring the resources and actors of a workplace, referenced by programs (environment)	Home kitchen with two cooks; small academic lab; theater; gym floor
Choice	A decision point whose options gate downstream steps (choice , choiceId)	Chicken or tofu; run the optional wash; rain plan; beginner or advanced set
Replicate and stagger	The same track or step repeated n times, in parallel, staggered by a delay, or serially (replicates , batch_size , stagger)	Three trays of cookies 10 min apart; twelve samples; three landings on one runway
Priority and flex	Ordering for planners when something must give, and permission for a step to stretch to fill a gap (priority , flex)	Drop the garnish before the main; let the rest interval fill the gap
Makespan, slack, critical chain	Total resolved time; how early a track finishes; the dependency chain that determines the end	Dinner is 4 h 40; the toast track has 6 min to spare; the turkey chain sets the finish
Manual gate and abort path	A step that waits for the executor’s signal; steps that run only if another is aborted (manual , onAbort)	“Start when guests are seated”; discard and re-plate if contaminated

Table 1: Concepts common to human-executed schedules and how Rhylthyme expresses them.